

GUJARAT TECHNOLOGICAL UNIVERSITY

Navyug Vidyabhavan Trust

C. K. Pithawalla College of Engineering and Technology, Surat

A

Project Report

On

(Student Management and Performance Analysis System)

Prepared as a part of the requirements for the subject of

Python for Data Science (BE05000231)

(PBL - Report_3)

B. E. III, Semester – V

(Computer Engineering)

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1. ABSTRACT

The **Student Management and Performance Analysis System** is a Python-based mini project developed to manage student records and analyze their academic performance. The main purpose of this project is to combine basic student management operations with Python for Data Science concepts.

The system uses **MySQL** to store student information. It provides different operations such as adding new student records, viewing existing records, searching for a particular student, updating student information and deleting records when required.

The project also performs data analysis on student academic records. **Pandas** is used to load and analyze the data, while **NumPy** is used for numerical operations. **SciPy** is used for statistical analysis. The system calculates values such as mean, median, minimum, maximum, standard deviation and mode.

For better understanding of the data, the project uses **Matplotlib and Seaborn** to create graphs. These graphs help in understanding student marks, subject performance and the relationship between attendance and academic performance.

The project also includes a machine learning component using **Scikit-learn**. A Decision Tree Classifier is used for Pass/Fail prediction. The current test produced an accuracy of **1.00 on the 7-record test set**.

As an additional topic beyond the prescribed syllabus, **Tkinter GUI development** is implemented. It provides a graphical interface through which the user can access the different features of the system.

The project uses a unique dataset containing **21 student records** and demonstrates the practical application of Python in database management, data analysis, visualization and machine learning.

2. INTRODUCTION

In educational institutions, student information such as marks, attendance and academic performance needs to be stored and managed properly. When student information is maintained manually, searching, updating and analyzing the information can take more time.

The **Student Management and Performance Analysis System** provides a simple solution to this problem. It combines student record management with data science techniques in one application.

The student information is stored in a MySQL database. Python is used to connect the application with the database and perform different operations on student records.

After collecting the data, the system performs data analysis using Python libraries. The analysis helps to identify the average marks, highest marks, lowest marks, attendance and other statistical values.

Data visualization is also included in the project. Instead of looking only at numerical values, graphs are used to represent the information visually. This makes it easier to understand differences in student performance.

The project also demonstrates a basic machine learning application. A Decision Tree model is used to perform Pass/Fail classification based on student performance information.

A Tkinter-based graphical interface is provided so that the user can operate the application without directly entering Python commands.

3. PROBLEM STATEMENT

Managing student records and analyzing academic performance are two important tasks in an educational environment. A simple record management system may store student information, but it may not provide useful analysis of the stored data.

The problem addressed by this project is to develop an application that can both **manage student records and analyze student performance**.

The system should be able to:

- Store student information in a database.
- Add new student records.
- Display existing student records.
- Search for a particular student.
- Update incorrect or changed information.
- Delete unnecessary records.
- Analyze student marks and attendance.
- Calculate statistical values.
- Display information through graphs.
- Perform basic Pass/Fail prediction using machine learning.
- Provide a simple graphical interface for the user.

The project therefore combines database operations and Python Data Science concepts in a single application.

4. OBJECTIVES

The main objectives of the project are:

4.1 Student Record Management

To develop a system that can store and manage student information using a MySQL database.

4.2 CRUD Operations

To implement the four basic database operations:

- Create
- Read
- Update
- Delete

These operations allow the user to manage student records easily.

4.3 Data Analysis

To analyze student marks and attendance using Python Data Science libraries such as Pandas, NumPy and SciPy.

4.4 Statistical Analysis

To calculate important statistical values such as mean, median, standard deviation, minimum, maximum and mode.

4.5 Data Visualization

To represent student performance using graphs created with Matplotlib and Seaborn.

4.6 Machine Learning

To implement a basic machine learning model that classifies students into Pass and Fail categories.

4.7 Graphical User Interface

To develop a simple GUI using Tkinter so that all major functions can be accessed easily.

4.8 Practical Application

To demonstrate how Python for Data Science concepts can be applied to a real-world type of student dataset.

5. TECHNOLOGIES AND LIBRARIES USED

The project uses several Python libraries and technologies.

Technology	Purpose
Python	Main programming language
MySQL	Database storage
MySQL Connector	Connect Python with MySQL
Tkinter	Graphical User Interface
NumPy	Numerical calculations
Pandas	Data analysis and manipulation
SciPy	Statistical calculations
Matplotlib	Graph creation
Seaborn	Statistical visualization
Scikit-learn	Machine learning
GitHub	Source code repository

Python

Python is used as the main programming language for developing the application and performing data science tasks.

MySQL

MySQL is used to store student information in a structured database. It supports **inserting, retrieving, updating, and deleting** student records.

Pandas

Pandas is used to handle the student dataset and perform data analysis.

NumPy

NumPy is used for numerical operations and handling numerical data efficiently.

SciPy

SciPy is used for statistical calculations such as standard deviation and mode.

Matplotlib and Seaborn

These libraries are used to create graphs and visual representations of the student data.

Scikit-learn

Scikit-learn is used to implement the Decision Tree machine learning model.

Tkinter

Tkinter is used to create the graphical interface of the application.

6. SYSTEM REQUIREMENTS

6.1 Hardware Requirements

The project does not require high-end hardware.

The minimum requirements are:

- Computer or laptop
- Minimum 4 GB RAM
- Keyboard
- Mouse
- Sufficient storage space

6.2 Software Requirements

The software requirements are:

- Windows/Linux operating system
- Python 3.x
- MySQL Server
- MySQL Connector for Python
- Required Python libraries
- Internet connection for downloading packages if required
- GitHub for project source-code storage

7. SYSTEM FEATURES

The application contains several major features.

7.1 Add Student

The Add Student feature allows the user to enter a new student's information and store it in the MySQL database.

7.2 View Students

The View Students feature displays the available student records from the database.

7.3 Search Student

The Search feature allows the user to find a student using the available student information.

7.4 Update Student

The Update feature allows existing student information to be modified.

7.5 Delete Student

The Delete feature allows an unwanted student record to be removed from the database.

7.6 Data Analysis

The Data Analysis feature loads the student data and calculates statistical information.

7.7 Visualization

The Visualization feature creates graphs that represent student performance.

7.8 ML Prediction

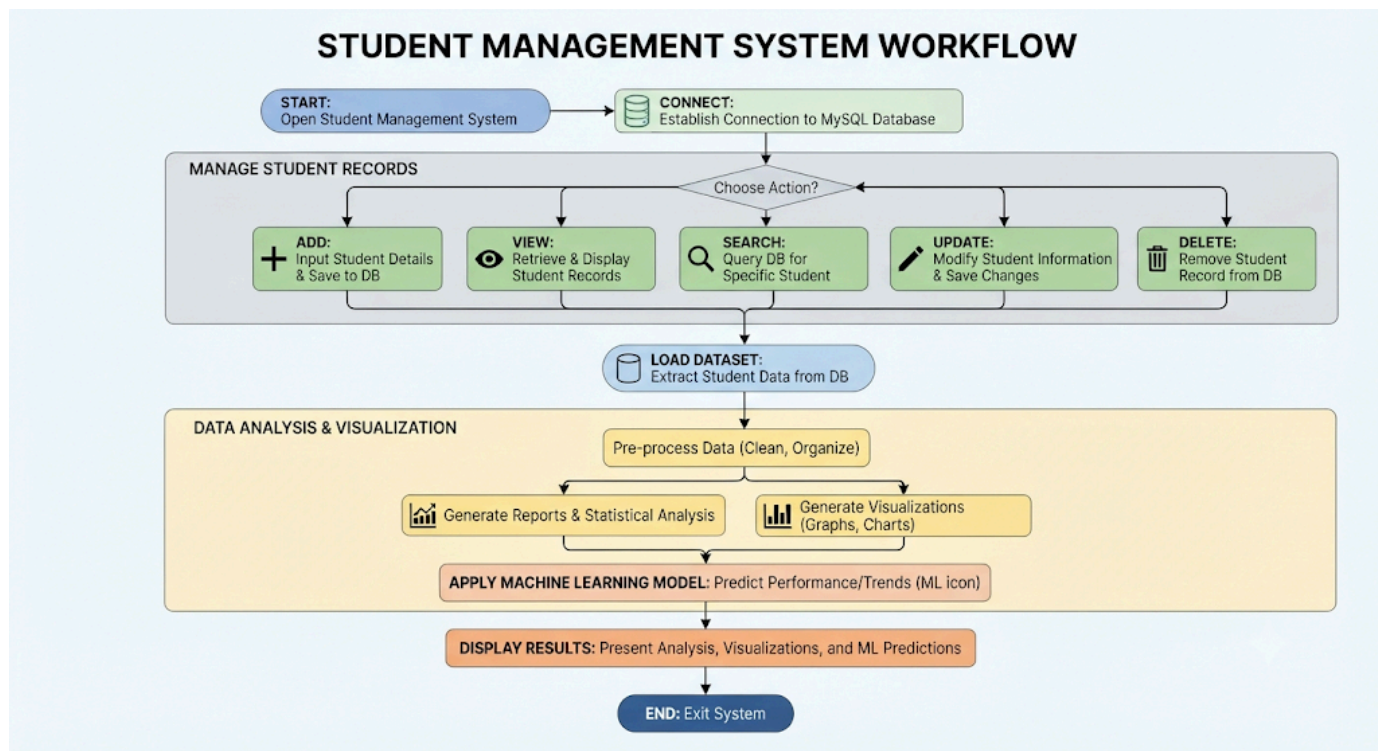
The ML Prediction feature applies a Decision Tree model to perform Pass/Fail classification.

7.9 GUI

The Tkinter interface provides buttons and input fields through which the user can operate the application.

8. SYSTEM WORKFLOW

The overall workflow of the project is:



Explanation

First, the application is started and connects to the MySQL database. The user can then perform different student management operations.

After the student data is available, the system can perform data analysis. Statistical values are calculated to understand the dataset.

The system then generates graphs to provide a visual representation of the data.

Finally, the machine learning module uses the student data to perform Pass/Fail classification and displays the result.

9. DATABASE DESIGN

The project uses **MySQL** as the database management system.

The database contains student information required for management and analysis.

The important fields include:

- Roll Number
- Student Name
- Mathematics Marks
- Python Marks
- Computer Networks Marks
- WAD Marks
- Average Marks
- Attendance

The database provides a structured way of storing the information.

The Python application connects to the database using the MySQL Connector library.

Database Working

Python Application



MySQL Connector



MySQL Database



Student Table



Student Records

The database makes it possible to permanently store the student records instead of keeping them only while the Python program is running.

10. CRUD OPERATIONS

CRUD stands for **Create, Read, Update and Delete**.

These are the basic operations performed on the student database.

Create

The Create operation is used to add a new student record to the database.

Read

The Read operation is used to retrieve and display existing student records.

Update

The Update operation is used when the information of an existing student needs to be changed.

Delete

The Delete operation removes a student record from the database.

These operations make the system useful as a basic student management application.

11. DATASET

The project uses a **unique dataset containing 21 student records**.

Each record contains academic information such as subject marks, average marks and attendance.

The dataset is used for both analysis and machine learning.

The student dataset allows the project to demonstrate different Data Science operations such as:

- Data loading
- Data selection
- Statistical analysis
- Data visualization
- Machine learning

The dataset is stored through the MySQL database and is accessed by the Python application.

ID	Roll No	Name	Branch	Semester	Email	Mobile	Math	Python	CN	WAD	Average
1	240090107070	Om Koradia	Computer Engineer	5	omskoradia@gmail.com	7698551768	78.0	80.0	82.0	75.0	78.75
2	240090107001	Aarav Patel	Computer Engineer	5	student001@example.com	9000000001	78.0	82.0	74.0	80.0	78.5
3	240090107002	Vivaan Shah	Computer Engineer	5	student002@example.com	9000000002	65.0	71.0	68.0	73.0	69.25
4	240090107003	Aditya Mehta	Computer Engineer	5	student003@example.com	9000000003	45.0	52.0	48.0	50.0	48.75
5	240090107004	Arjun Desai	Computer Engineer	5	student004@example.com	9000000004	88.0	91.0	85.0	89.0	88.25
6	240090107005	Rohan Shah	Computer Engineer	5	student005@example.com	9000000005	35.0	42.0	38.0	40.0	38.75
7	240090107006	Krish Patel	Computer Engineer	5	student006@example.com	9000000006	72.0	68.0	75.0	70.0	71.25
8	240090107007	Dhruv Joshi	Computer Engineer	5	student007@example.com	9000000007	55.0	61.0	58.0	63.0	59.25
9	240090107008	Yash Mehta	Computer Engineer	5	student008@example.com	9000000008	92.0	87.0	90.0	94.0	90.75
10	240090107009	Dev Shah	Computer Engineer	5	student009@example.com	9000000009	38.0	35.0	42.0	39.0	38.5
11	240090107010	Parth Patel	Computer Engineer	5	student010@example.com	9000000010	67.0	74.0	70.0	72.0	70.75
12	240090107011	Harsh Desai	Computer Engineer	5	student011@example.com	9000000011	81.0	77.0	83.0	79.0	80.0
13	240090107012	Meet Shah	Computer Engineer	5	student012@example.com	9000000012	49.0	45.0	52.0	48.0	48.5
14	240090107013	Manav Patel	Computer Engineer	5	student013@example.com	9000000013	73.0	79.0	71.0	76.0	74.75
15	240090107014	Reyansh Mehta	Computer Engineer	5	student014@example.com	9000000014	58.0	54.0	61.0	57.0	57.5
16	240090107015	Jay Shah	Computer Engineer	5	student015@example.com	9000000015	89.0	84.0	91.0	87.0	87.75
17	240090107016	Kunal Patel	Computer Engineer	5	student016@example.com	9000000016	42.0	39.0	45.0	44.0	42.5
18	240090107017	Ansh Desai	Computer Engineer	5	student017@example.com	9000000017	76.0	72.0	78.0	75.0	75.25
19	240090107018	Devansh Shah	Computer Engineer	5	student018@example.com	9000000018	63.0	67.0	59.0	65.0	63.5
20	240090107019	Om Patel	Computer Engineer	5	student019@example.com	9000000019	31.0	37.0	34.0	36.0	34.5
21	240090107020	Aayush Mehta	Computer Engineer	5	student020@example.com	9000000020	84.0	88.0	82.0	86.0	85.0

12. DATA ANALYSIS

Data analysis is an important part of the project. It is used to understand the academic performance of the students.

The project uses **Pandas** for handling the data and **NumPy and SciPy** for numerical and statistical calculations.

The following values are calculated:

- Number of records
- Mean
- Median
- Minimum
- Maximum
- Standard deviation
- Mode
- Average attendance

Obtained Results

Parameter	Result
Number of Students	21
Average Mathematics Marks	64.71
Average Python Marks	65.95
Average CN Marks	66.00
Average WAD Marks	66.57
Overall Average Marks	65.81
Average Attendance	82.81%
Highest Average Marks	90.75
Lowest Average Marks	34.50
Standard Deviation	17.96
Mode	34.50

```

=====
STUDENT DATA ANALYSIS
=====

Descriptive Statistics:
count      math_marks  python_marks  ...  average_marks  attendance
mean       64.714286   65.952381    ...   65.809524     82.809524
std        18.905404   18.246852    ...   17.957920     8.189133
min        31.000000   35.000000    ...   34.500000     68.000000
25%        49.000000   52.000000    ...   48.750000     77.000000
50%        67.000000   71.000000    ...   70.750000     82.000000
75%        78.000000   80.000000    ...   78.750000     89.000000
max        92.000000   91.000000    ...   90.750000     96.000000

[8 rows x 6 columns]

Mean:
math_marks      64.714286
python_marks    65.952381
cn_marks        66.000000
wad_marks       66.571429
average_marks   65.809524
attendance      82.809524
dtype: float64

Median:
math_marks      67.00
python_marks    71.00
cn_marks        70.00
wad_marks       72.00
average_marks   70.75
attendance      82.00
dtype: float64

Highest Average Marks:
90.75

Lowest Average Marks:
34.5

Average Attendance:
82.80952380952381

SciPy Statistical Analysis:
Standard Deviation of Average Marks:
17.957920390788704
Mode of Average Marks:
34.5

```

Interpretation

- The overall average marks are **65.81**.
- Average attendance is **82.81%**.
- **WAD** has the highest subject average (**66.57**), while Mathematics has the lowest (**64.71**).
- Highest individual average is **90.75**, and lowest is **34.50**.
- Standard deviation is **17.96**, showing variation in students' average marks.

13. DATA VISUALIZATION

Data visualization is used to represent numerical information using graphs.

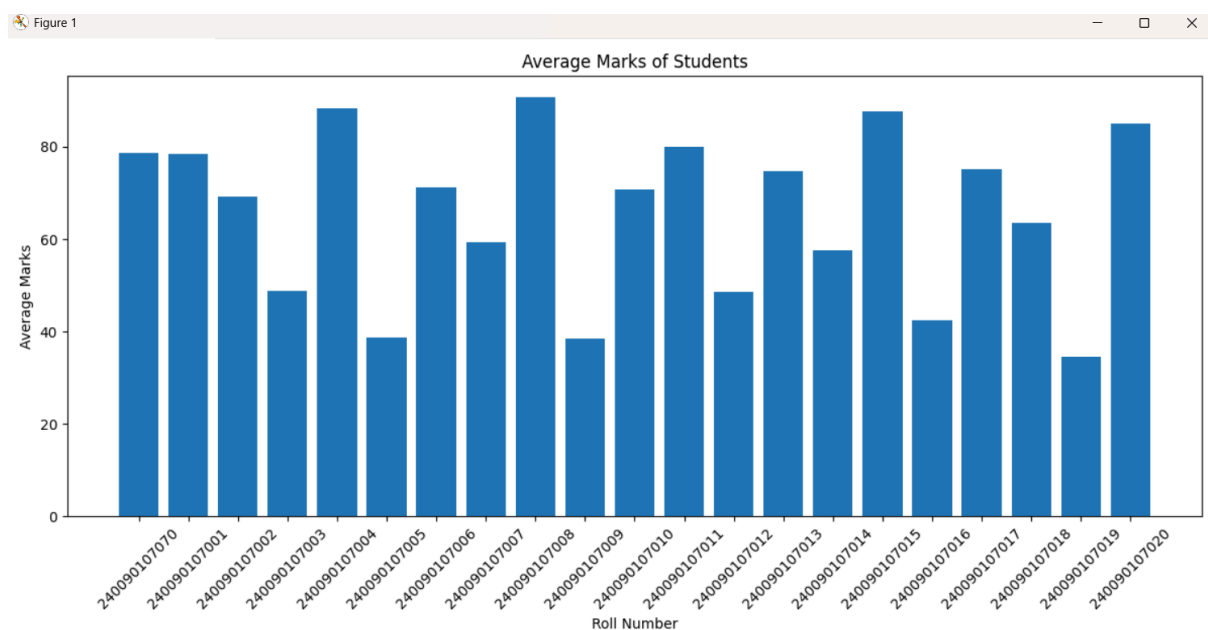
The project uses **Matplotlib and Seaborn** for visualization.

Graphs provide an easier way to understand the student data because patterns and differences can be seen visually.

The project generates three visualizations.

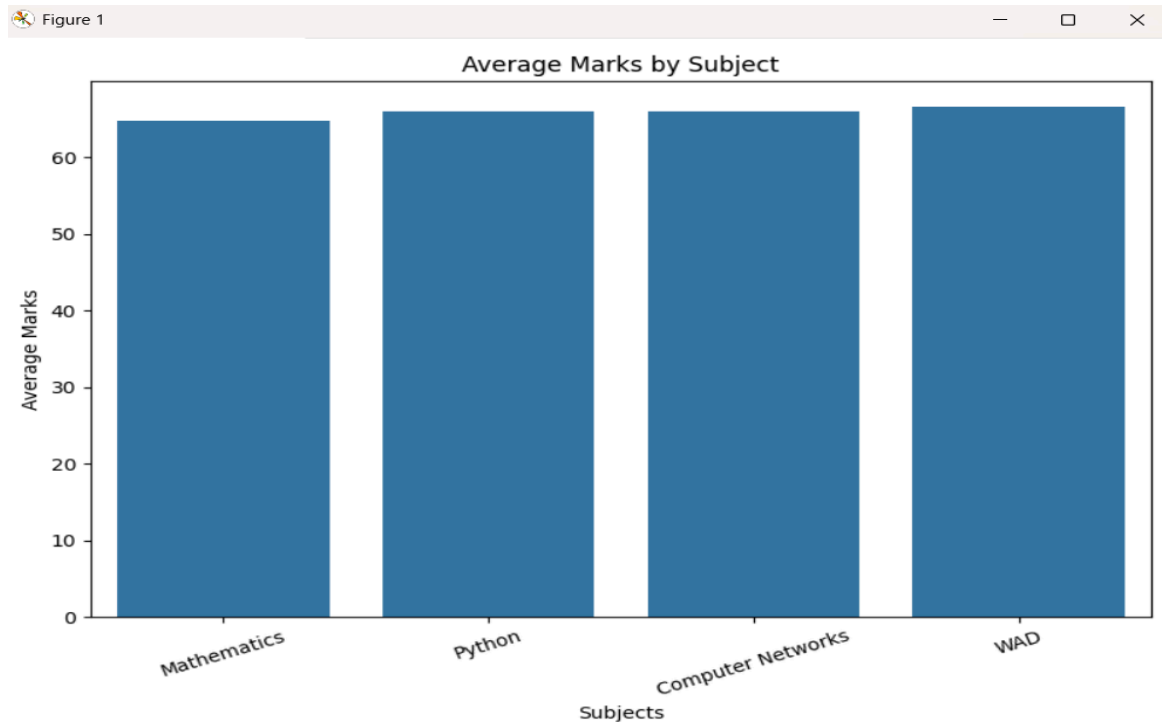
Visualization 1 – Student Average Performance

This graph represents student average marks and helps compare the academic performance of different students.



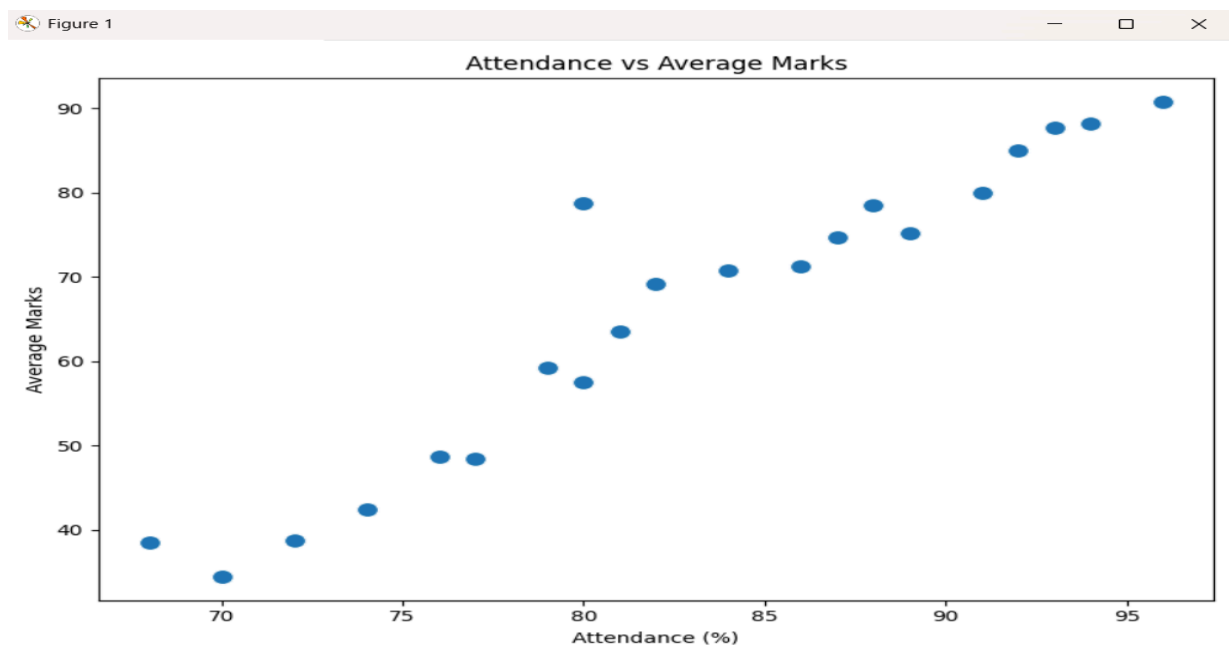
Visualization 2 – Subject-wise Performance

This graph compares the average performance across the different subjects.



Visualization 3 – Attendance and Performance

This graph represents the relationship between student attendance and academic performance.



Importance of Visualization

Visualization makes numerical data easier to understand. Instead of checking every value individually, the user can observe the graph and identify differences in performance.

14. MACHINE LEARNING

Machine learning is included in the project to demonstrate how student data can be used for classification.

The project uses **Scikit-learn** and a **Decision Tree Classifier**.

The model performs Pass/Fail classification based on the student performance information used by the application.

Steps Used

1. Student data is loaded.
2. Required features are selected.
3. Data is divided into training and testing sets.
4. A Decision Tree model is created.
5. The model is trained using the training data.
6. Predictions are made using the test data.
7. Accuracy and classification metrics are calculated.

Result

The current dataset contains:

Total records: 21

Test records: 7

Accuracy: 1.00

Therefore, the current test run produced:

Accuracy = 100%

Classification Report

Class	Precision	Recall	F1-Score
0	1.00	1.00	1.00
1	1.00	1.00	1.00

Observation

The model correctly classified all 7 records in the current test set.

Since the dataset is small, the result should be considered as the result of this particular test split. A larger dataset would be required to properly evaluate how the model performs on new student records.

```

=====
MACHINE LEARNING
=====

Number of Records: 21

Accuracy: 1.0

Classification Report:

```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1
1	1.00	1.00	1.00	6
accuracy			1.00	7
macro avg	1.00	1.00	1.00	7
weighted avg	1.00	1.00	1.00	7

15. BEYOND SYLLABUS TOPIC

Tkinter GUI Development

The project includes **Tkinter GUI development** as the additional topic beyond the syllabus provided for the project.

Tkinter is used to create the graphical user interface of the Student Management and Performance Analysis System.

Instead of requiring the user to enter Python commands, the application provides buttons and input fields.

The GUI provides access to:

- Add Student
- View Students
- Search Student
- Update Student
- Delete Student
- Data Analysis
- Visualization
- ML Prediction

Benefits of Tkinter GUI

- Simple to use.
- Easy to understand.
- Provides a graphical interface.
- Reduces the need for command-line input.
- Combines all project features in one window.

Student Management & Performance Analysis System

Roll No Name Branch Semester
Email Mobile Math Marks Python Marks
CN Marks WAD Marks Attendance %

Add Student

Update Student

Delete Student

Clear

View Students

Data Analysis

Visualization

ML Prediction

Search Roll No / Name:

Search

Show All

ID	Roll No	Name	Branch	Semester	Email	Mobile	Math	Python	CN	WAD	Average
1	240090107070	Om Koradia	Computer Engineer	5	omskoradia@gmail.com	7698551768	78.0	80.0	82.0	75.0	78.75
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21	240090107020	Aayush Mehta	Computer Engineer	5	student020@example.com	9000000020	84.0	88.0	82.0	86.0	85.0

16. IMPLEMENTATION

The project was implemented in multiple stages.

Step 1 – Database Creation

A MySQL database and student table were created to store the student information.

Step 2 – Database Connection

Python was connected to MySQL using the MySQL Connector library.

Step 3 – Student Management

CRUD functions were implemented for adding, viewing, searching, updating and deleting student records.

Step 4 – Data Analysis

Pandas was used to load and analyze the student data. NumPy and SciPy were used for numerical and statistical calculations.

Step 5 – Visualization

Matplotlib and Seaborn were used to generate graphs.

Step 6 – Machine Learning

A Decision Tree Classifier from Scikit-learn was implemented for Pass/Fail classification.

Step 7 – GUI

Tkinter was used to combine the different functions into a graphical application.

17. IMPORTANT CODE

The following code sections show the main implementation of the project.

17.1 Database Connection

Code

```
connection=mysql.connector.connect(  
    host="localhost",  
    user="root",  
    password=MYSQL_PASSWORD,  
    database="student_management"  
)
```

17.2 Data Analysis

Code

```
df=pd.DataFrame(data)  
print(df.describe())  
  
print("Mean:")  
print(df.mean(numeric_only=True))  
  
print("Median:")  
print(df.median(numeric_only=True))
```

17.3 Machine Learning

Code

```
model=DecisionTreeClassifier(random_state=42)  
model.fit(X_train, y_train)  
  
prediction=model.predict(X_test)  
accuracy=accuracy_score(y_test, prediction)  
print("Accuracy:", accuracy)
```

17.4 Tkinter GUI

Code

```
import tkinter as tk  
  
root = tk.Tk()  
  
root.title("Student Management System")  
  
root.mainloop()
```

18. OUTPUT

The project was tested successfully and produced the required outputs.

Student Management Output

The application successfully performed:

- Add Student
- View Students
- Search Student
- Update Student
- Delete Student

Data Analysis Output

The system successfully calculated statistical information from the 21 student records.

Visualization Output

The application successfully generated three graphs.

Machine Learning Output

The Decision Tree model successfully generated the classification result with an accuracy of **1.00 for the current test set.**

GUI Output

The Tkinter GUI successfully provided access to all major project features.

19. RESULTS AND OBSERVATIONS

The developed system successfully combines database management with Python Data Science concepts.

The main observations are:

1. Student records can be managed using MySQL.
2. CRUD operations work successfully.
3. Statistical analysis can be performed on the student dataset.
4. The average marks and attendance can be calculated.
5. Graphs provide a visual understanding of student performance.
6. Machine learning can be used for Pass/Fail classification.
7. Tkinter provides a simple graphical interface for the complete application.
8. The current project successfully works with a dataset of 21 student records.

20. ADVANTAGES

The main advantages of the system are:

Easy Record Management

Student information can be added, viewed, searched, updated and deleted easily.

Database Storage

Student data is stored in MySQL in a structured form.

Data Analysis

The system provides useful statistical information about student performance.

Visualization

Graphs make it easier to understand student data.

Machine Learning

The project demonstrates the use of machine learning for Pass/Fail classification.

User-Friendly Interface

Tkinter provides a simple graphical interface.

Integrated System

Database management, data analysis, visualization and machine learning are available in one application.

21. LIMITATIONS

Although the project provides the required functionality, it has some limitations.

1. The current dataset contains only 21 records.
2. The machine learning model is trained using a small dataset.
3. The current project is designed as a mini project rather than a large institutional system.
4. More student data would be required for stronger machine learning evaluation.
5. The system currently focuses on a limited number of academic parameters.

22. FUTURE SCOPE

The project can be improved in the future by adding more features.

Larger Dataset

More student records can be added to improve data analysis and machine learning evaluation.

More Subjects

Additional subjects and academic parameters can be included.

Login System

A login system can be added so that only authorized users can access the application.

Student Reports

The application can generate individual student performance reports.

More Machine Learning Models

Other classification algorithms can be tested and compared.

Web Application

The desktop application can later be converted into a web-based system.

Advanced Analytics

Additional statistical analysis and performance prediction features can be added.

23. CONCLUSION

The **Student Management and Performance Analysis System** was successfully developed using Python.

The project provides basic student record management using **MySQL** and performs data analysis using **Pandas, NumPy and SciPy**. **Matplotlib and Seaborn** are used to visualize the student data.

A **Decision Tree Classifier** from Scikit-learn is used for Pass/Fail classification. The current test run produced an accuracy of **1.00 on the 7-record test set**.

The project also implements **Tkinter GUI development as the beyond-syllabus topic**, providing a simple graphical interface for accessing the system's features.

Overall, the project demonstrates the practical use of Python for **database management, data analysis, statistical analysis, visualization and machine learning**. It provides a simple example of how Data Science concepts can be applied to student academic data.

24. REFERENCES

Python Documentation – <https://docs.python.org/3/>

MySQL Documentation – <https://dev.mysql.com/doc/>

NumPy Documentation – <https://numpy.org/doc/>

Pandas Documentation – <https://pandas.pydata.org/docs/>

SciPy Documentation – <https://docs.scipy.org/doc/scipy/>

Matplotlib Documentation – <https://matplotlib.org/stable/>

Seaborn Documentation – <https://seaborn.pydata.org/>

Scikit-learn Documentation – <https://scikit-learn.org/stable/>

Tkinter Documentation – <https://docs.python.org/3/library/tkinter.html>

Project GitHub Repository – <https://github.com/OmKoradia/student-management-system>

25. APPENDIX

Appendix A – Complete Python Source Code

[Click here to view the complete Python source code](#)

Appendix B – Database Source Code

[Click here to view the complete MySQL database code](#)